2025. 금호중3. 수학. 1중간

$\boxed{1}$ ③

① 64의 제곱근 : ± 8

② $\sqrt{(-3)^2} = 3$

④ $\sqrt{25}$의 제곱근 : $\pm\sqrt{5}$

⑤ $(-6)^2$의 제곱근 : ± 6

$\boxed{2}$ ④

④ $(-\sqrt{6})^2 = 6$

$\boxed{3}$ ③ $\boxed{4}$

π, $\sqrt{11}$, $2+\sqrt{3}$, $\sqrt{1000}$ 4개

$\boxed{4}$ ① $\boxed{23}$

$n=4$ 일때 $\sqrt{20-4} = \sqrt{16} = 4$, $A=4$

$n=19$ 일때, $\sqrt{20-19} = \sqrt{1} = 1$, $B=19$

$A+B = 23$

$\boxed{5}$ ⑤ $\boxed{4\sqrt{2}}$

$\overline{AC}^2 = \overline{AB}^2 + \overline{BC}^2 = 4+4 = 8$, $\overline{AC} = 2\sqrt{2}$

$P: 2+2\sqrt{2}$, $Q: 2-2\sqrt{2}$

$P-Q = (2+2\sqrt{2}) - (2-2\sqrt{2}) = 4\sqrt{2}$

$\boxed{6}$ ②

$(5-\sqrt{6}) - (2-\sqrt{6}) = 3 > 0$

$\therefore 5-\sqrt{6} > 2-\sqrt{6}$

$\boxed{7}$ ④ $\frac{6}{\sqrt{24}} = \frac{\sqrt{6}}{2}$

$\boxed{8}$ ①

① $\sqrt{10}$ ② $\sqrt{6}$ ③ $2\sqrt{2}$ ④ $\sqrt{6}$ ⑤ 3

$\boxed{9}$ ④ $\boxed{2\sqrt{3}}$

$\sqrt{32} = 4\sqrt{2}$, $a=4$

$4\sqrt{3} = \sqrt{48}$, $b=48$

$\sqrt{\frac{b}{a}} = \sqrt{\frac{48}{4}} = \sqrt{12} = 2\sqrt{3}$

$\boxed{10}$ ⑤ $\boxed{42}$

$f(1) = f(2) = f(3) = 1$

$f(4) = f(5) = \cdots = f(8) = 2$

$f(9) = f(10) = \cdots = f(15) = 3$

$f(16) = f(17) = 4$

$1\times3 + 2\times5 + 3\times21 + 4\times2 = 42$

$\boxed{11}$ ① $\boxed{\sqrt{2}+3\sqrt{6}}$

$\boxed{12}$ ② $\boxed{-\sqrt{15}+9\sqrt{3}}$

$\boxed{13}$ ② $\boxed{-1}$

$a=2$, $b=3$, $a-b=-1$

$\boxed{14}$ ④ $\boxed{18}$

$(2x+a)(3x+5) = 6x^2 + (10+3a)x + 5a$

$5a=10$, $a=2$, $b=16$, $a+b=18$

$\boxed{15}$ ③

$(x+5)^2 = x^2 + 10x + 25$

16 ① $\boxed{-\frac{35}{4}}$

$2p+1=6$ 또는 $2p+1=-6$

$p=\frac{5}{2}$ 또는 $p=-\frac{7}{2}$

$\frac{5}{2}\times\left(-\frac{7}{2}\right)=-\frac{35}{4}$

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17 ⑤ $\boxed{7}$

$a=5,\ b=2,\ a+b=7$

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18 ③

① $x^2-2x=x(x-2)$

② $x^2-4x+4=(x-2)^2$

③ $x^2+x-2=(x+2)(x-1)$

④ $x^2+x-6=(x-2)(x+3)$

⑤ $x^2-4=(x+2)(x-2)$

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19 ② $\boxed{-2x+3}$

$\sqrt{x^2+4x+4}+\sqrt{9x^2-6x+1}$

$=\sqrt{(x+2)^2}+\sqrt{(3x-1)^2}$

$=\boxed{+}(x+2)+\boxed{-}(3x-1)=-2x+3$

$x>-2,\quad x+2>0$

$x<\frac{1}{3},\quad 3x-1<0$

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20 ⑤ $\boxed{6}$

$x^2y+xy^2-x-y=16$

$xy\underbrace{(x+y)}_{4}-\underbrace{(x+y)}_{4}=16$

$4xy=20,\quad xy=5$

$x^2+y^2=(x+y)^2-2xy$

$=16-10=6$

서1 (1) $\boxed{\pm6}$, (2) $\boxed{\pm3}$

(1) 36의 제곱근은 ±6 이다.

(2) $\sqrt{81}$의 제곱근은 ±3 이다.

서2 $\boxed{\frac{1}{5}}$

$6-\sqrt{5}$의 정수부분 : 3

$6-\sqrt{5}$의 소수부분 : $3-\sqrt{5}=x$

$\frac{3+x}{3-x}=\frac{3+(3-\sqrt{5})}{3-(3-\sqrt{5})}=\frac{6-\sqrt{5}}{\sqrt{5}}$

$=-1+\frac{6}{5}\sqrt{5},\quad A=-1,\quad B=\frac{6}{5}$

$A+B=\frac{1}{5}$

서3 (1) $\boxed{4x+4y=44}$

(2) $\boxed{x^2+y^2=65}$ (3) $\boxed{448}$

$x^2+y^2=(x+y)^2-2xy,\quad 65=121-2xy$

$xy=28,\quad 4x\times4y=16xy=448$

서4 (1) $\boxed{(2x+1)(x-3),\ (x-3)(x-5)}$

(2) $\boxed{x-3}$